

Name:

in we'll take the norma

Points: /20

Databasutveckling och lagring — Final Exam: RETAKE II

2025-03-06

Instructions

1. **No computers, no phones, no notes**
2. **No talking with other people in the room**
3. In case point 1 or 2 are broken, you get automatically **0 points** on the test.
4. Write your name on top of the page (do not fill in the *Points* box!)
5. For each question, you'll need to answer the question by crossing with an **X** the **right** answer (multiple choice questions) or write your answer in the empty lines below the question (open questions).
6. Once you're done:
 - Double-check your answers
 - Hand over your test to me
 - Leave the room

NB: While inside the classroom, rule 1 and 2 are still valid, even if you've already handed over your answers to me. If you want to discuss your answers with a classmate, or to check an answer on your phone, do that **outside the classroom**.

NB: Make sure to use the restroom, fill our water bottle, grab a coffee, etc... **before** we begin the exam, as you're not allowed to leave the classroom without handing over your answers.

Point system

There are **10 questions**, worth **2 points** each, for a total of maximum **20 points**.

There are no partial points

You either give the **complete, correct answer**, and get 2 points, or get 0 points.

Grading

Points	Grade
0 - 12	IG
13 - 19	G
20	VG

READ EVERYTHING CAREFULLY!

1 Normalization

Given the following schema of a SQLite database:

Listing 1: The complete schema for the DB

```
PRAGMA foreign_keys = ON;

CREATE TABLE IF NOT EXISTS customer (
  customer_id INTEGER PRIMARY KEY,
  full_name VARCHAR(100) NOT NULL,
  email VARCHAR(100) UNIQUE NOT NULL,
  postal_code VARCHAR(5) NOT NULL,
  city VARCHAR(50) NOT NULL
);

CREATE TABLE IF NOT EXISTS orders (
  order_id INTEGER PRIMARY KEY,
  order_date TEXT NOT NULL,
  customer_id INTEGER NOT NULL,
  FOREIGN KEY (customer_id) REFERENCES customer(customer_id)
);

CREATE TABLE IF NOT EXISTS products (
  product_id INTEGER PRIMARY KEY,
  product_name TEXT NOT NULL,
  category TEXT,
  unit_price FLOAT NOT NULL
);

CREATE TABLE IF NOT EXISTS order_items (
  order_id INTEGER NOT NULL,
  product_id INTEGER NOT NULL,
  quantity INTEGER NOT NULL,
  product_name TEXT NOT NULL,
  PRIMARY KEY (order_id, product_id),
  FOREIGN KEY (order_id) REFERENCES orders(order_id),
  FOREIGN KEY (product_id) REFERENCES products(product_id)
);
```

Point out and explain **one** normalization problem present in the schema.


 Ett problem är att tabellen order_items
 inte har en Primary key vilket
 bryter emot 1NF

- 2 Explain how you would solve the normalization problem you described above.

You don't need to write a SQL query, just explain how you would fix it.

Genom att vi normaliserar de. Så
att vi kan undvika anomalies så som
Insertion, Update, Deletion. Inga partiella
beroenden, Inga transitiva beroenden.

- 3 In this course we've labbed with a database that contained the data for a music store. What was the name of the database, and which RDBMS did it use?

Name: Music

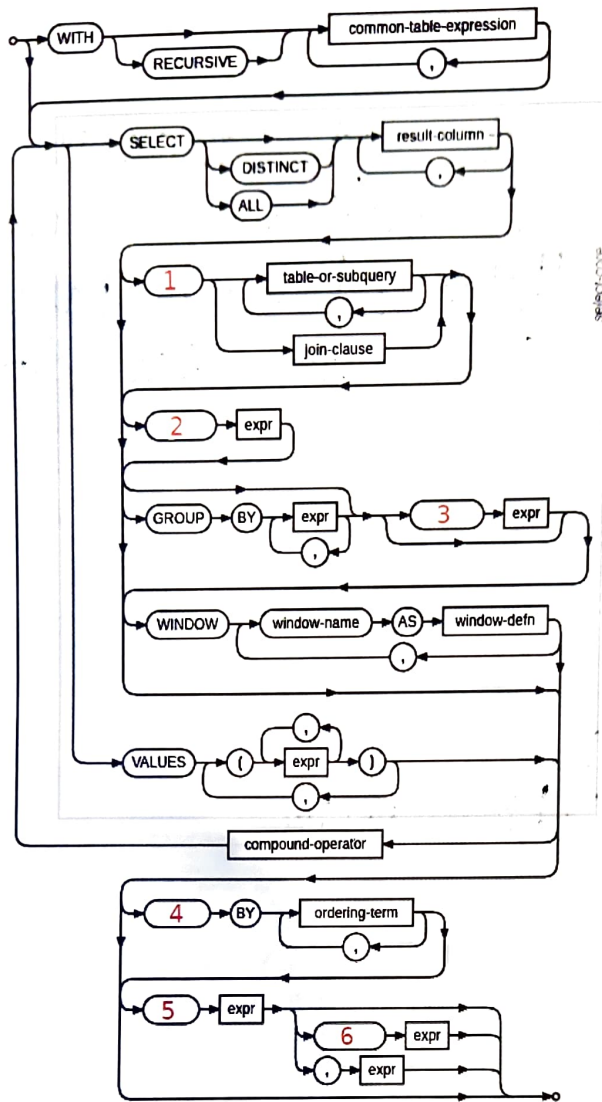
RDBMS: MariaDB SQLite

- 4 In this course, we've labbed with a database that contained the data for a movie rental store. What was the name of the database, and which RDBMS did it use?

Name: ~~Movies and Shows~~ Shows

RDBMS: MariaDB

5 Fill the blanks



1: From

2: ~~Where~~ Where

3: Having

4: ORDER

5: ~~Like~~ Like

6: Limit

6 What's the difference between WHERE and HAVING in a SELECT query?

Where hjälper oss att filtrera ut data. Having hjälper oss att söka mer efter specifika värden i en tabell

7 What's the difference between 'JOIN...USING' and 'JOIN...ON'?

Join using innebär att den är iakt

Join on betyder att den är påslagen

8 We can always replace 'JOIN...ON' with 'JOIN...USING'

1. True

2. ~~False~~

9 We can always replace 'JOIN...USING' with 'JOIN...ON'

1. True

2. ~~False~~

10 What is a subquery?

Innebär ett sätt för oss
att kunna "kommunicera" med
databaser. För att kunna modifiera.
Inserta, Selecta, radera osv ...